Event-Driven DevOps: Automating Azure with Event Grid System Topics

Hans Muns

Azure User Group
Norway

I like interactive sessions, please ask questions anytime

Agenda

- The spark ✨
- Down the rabbit hole 🐰
- Explore even further 🔭
- Go experiment 🧪

Hans Muns

Freelance Software Engineer @ Backnook, NL

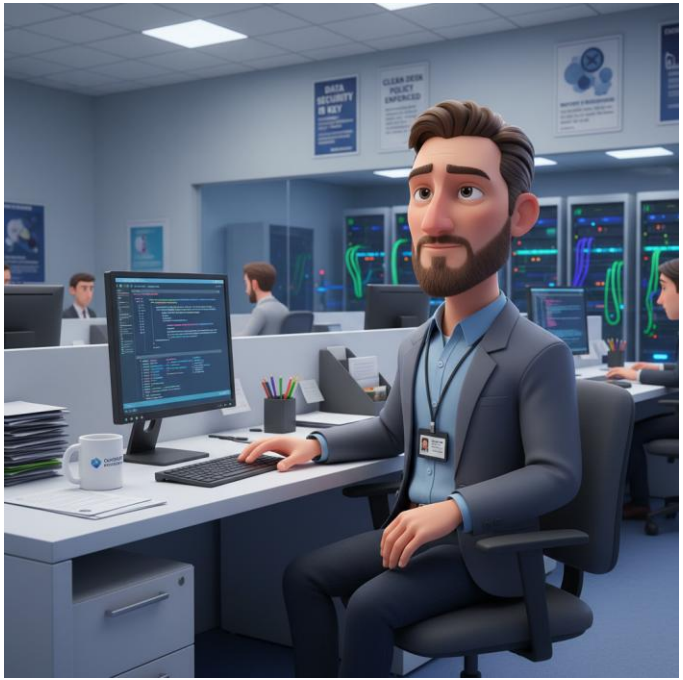
I do IT stuff

hans@backnook.nl

/in/hans-muns-78a50b197

The Spark ✨

Where did this all begin?



The Spark ✨

Where did this all begin?

Deploy Bicep file 7s

```
1  ▶ Run az deployment group create \
10
10 WARNING: /home/runner/work/policy-demo-invalid/policy-demo-invalid/infra/main.bicep(6,7) : Warning no-unused-params: Parameter "test" is declared but never used. [https://aka.ms/bicep/linter-diagnostics#no-unused-params]
12 ERROR: {"code": "InvalidTemplateDeployment", "message": "The template deployment failed because of policy violation. Please see details for more information."}
13 Inner Errors:
14 {"code": "RequestDisallowedByPolicy", "target": "myUserAssignedIdentity", "message": "Resource 'myUserAssignedIdentity' was disallowed by policy. Policy identifiers: '[{\"policyAssignment\":{\"name\":\"RequireEnvironmentTag\",\"id\":\"/subscriptions/***/providers/Microsoft.Authorization/policyAssignments/d894c3968c15464db1f2277a\"},\"policyDefinition\":{\"name\":\"Require a tag on resources\",\"id\":\"/providers/Microsoft.Authorization/policyDefinitions/871b6d14-10aa-478d-b590-94f262ecfa99\",\"version\":\"1.0.1\"},\"policySetDefinition\":{\"name\":\"RequireEnvironmentTag\",\"id\":\"/subscriptions/***/providers/Microsoft.Authorization/policySetDefinitions/91b871a292cb47749342ecaf\",\"version\":\"1.0.0\"}}]'.", "additionalInfo": [{"type": "PolicyViolation", "info": {"evaluationDetails": {"evaluatedExpressions": [{"result": "True", "expressionKind": "Field", "expression": "tags[Environment]", "path": "tags[Environment]", "targetValue": "false", "operator": "Exists"}]}, "policyDefinitionId": "/providers/Microsoft
16 Error: Process completed with exit code 1.
```

The Spark ✨

Where did this all begin?

```
d/policy-demo-invalid/infra/main.bicep(6,7) : Warning no-unused-params: Parameter "test" is declared but never used. [ht

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ions/***/providers/Microsoft.Authorization/policyAssignments/d894c3968c15464db1f2277a\"},\"policyDefinition\":{\"name\"
itions/871b6d14-10aa-478d-b590-94f262ecfa99\",\"version\":\"1.0.1\"},\"policySetDefinition\":{\"name\":\"RequireEnvironm
finitions/91b871a292cb47749342ecaf\",\"version\":\"1.0.0\"}}]'.", "additionalInfo": [{"type": "PolicyViolation", "info"
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Down the rabbit hole

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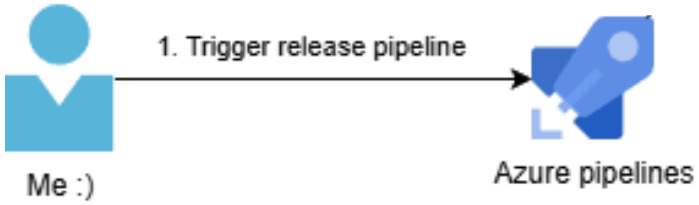

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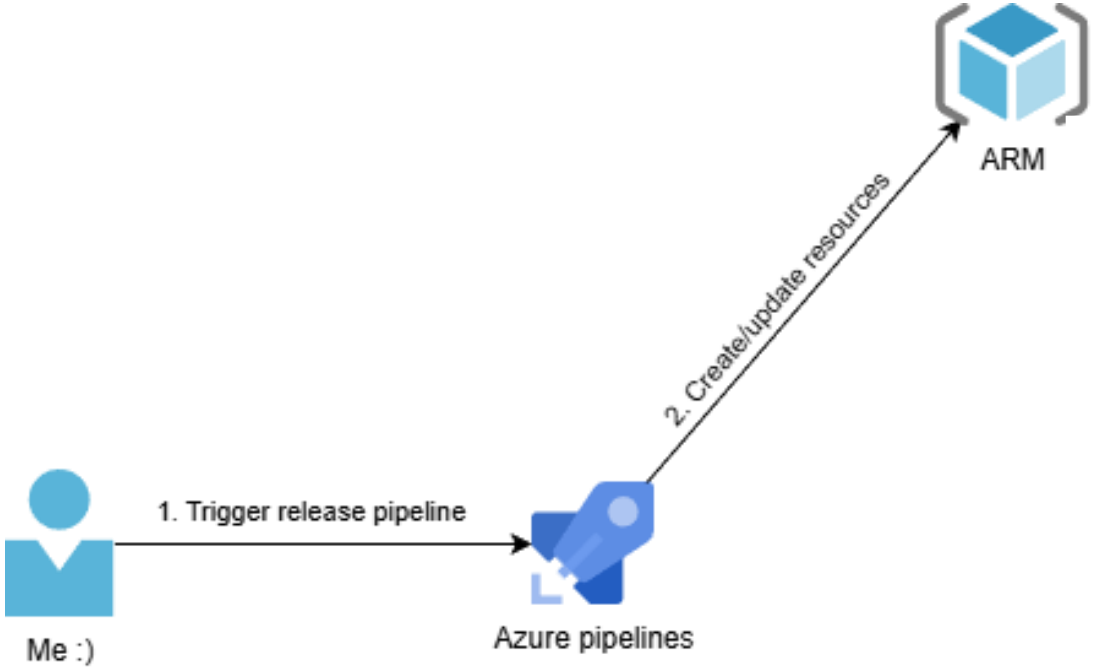

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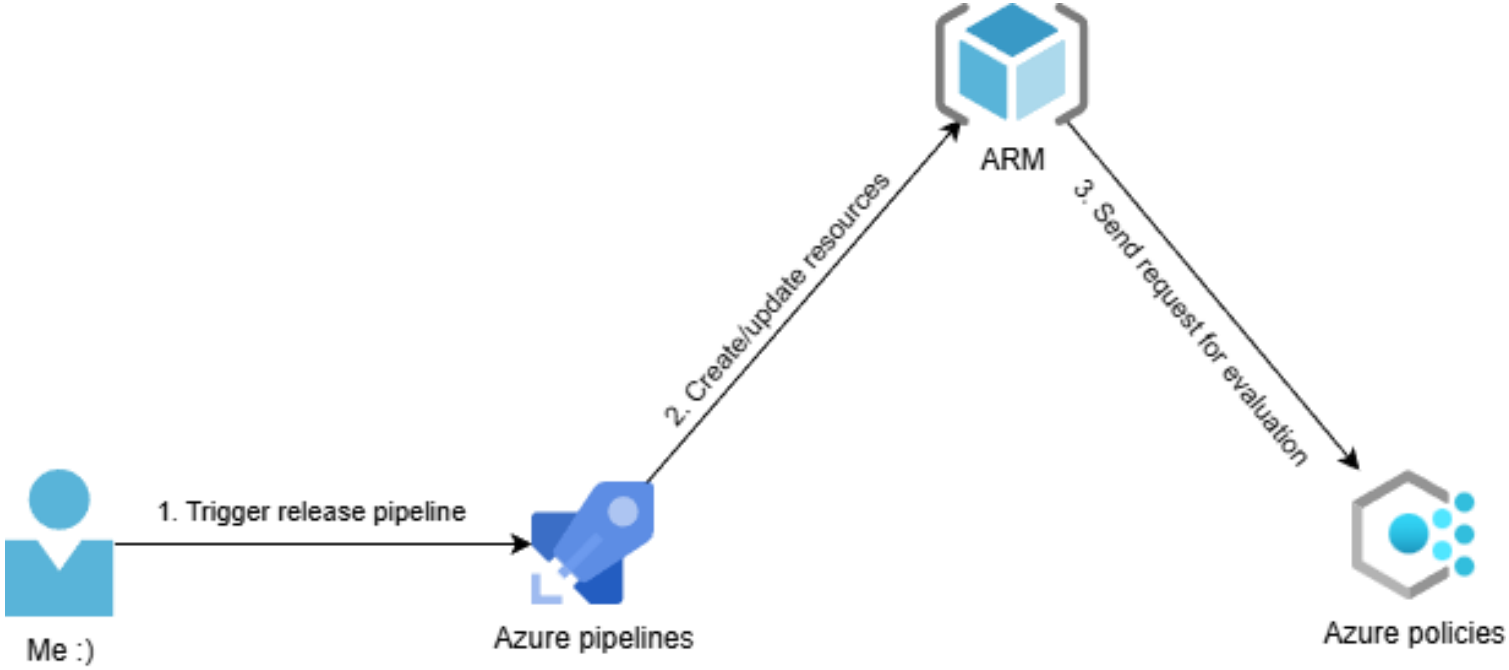

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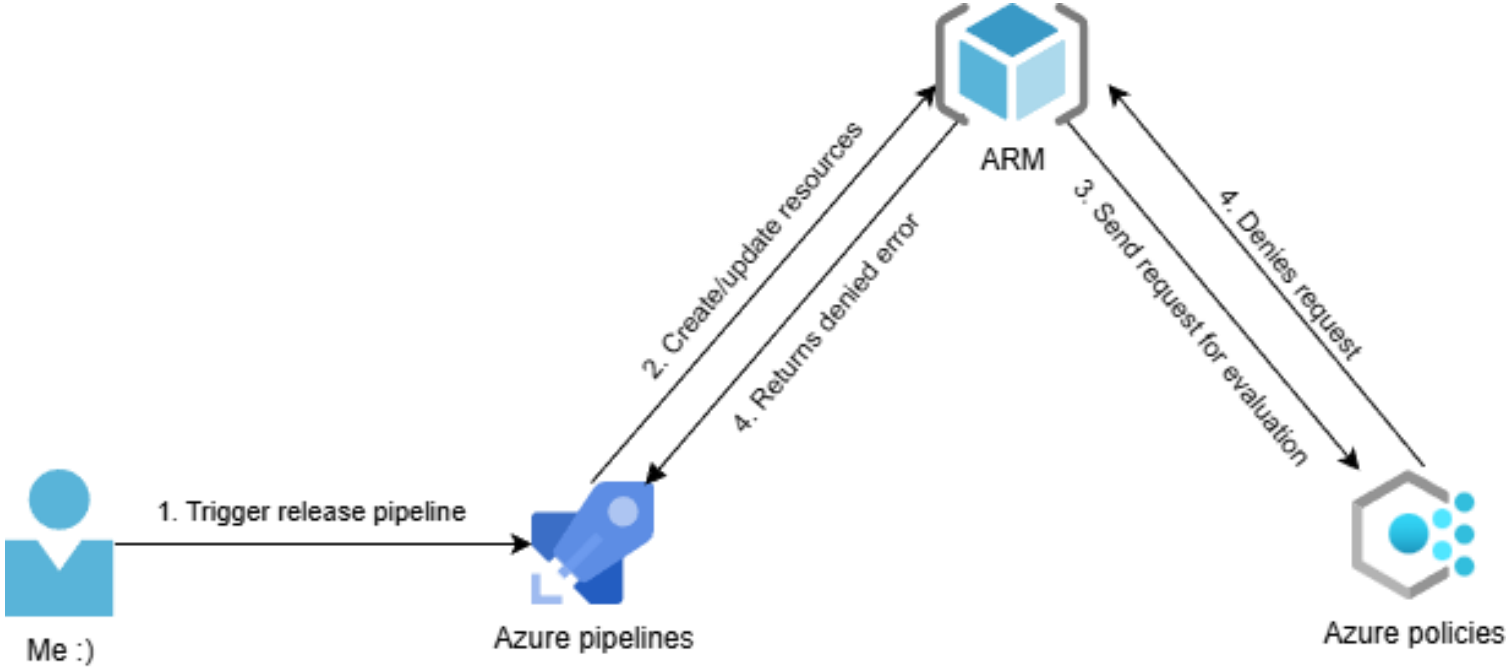


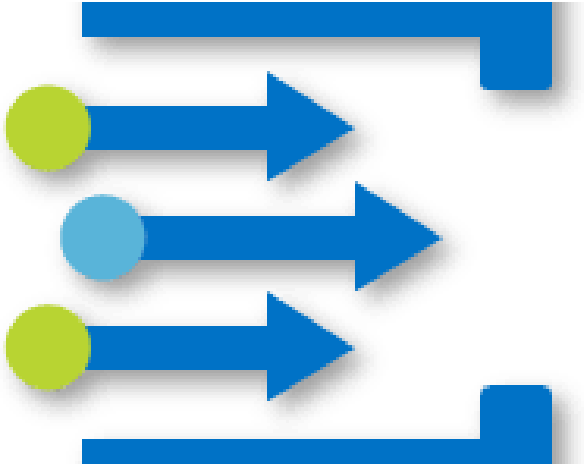
Down the rabbit hole 🐰

How can we automate this?

How can I get notified of these errors automatically?

Down the rabbit hole

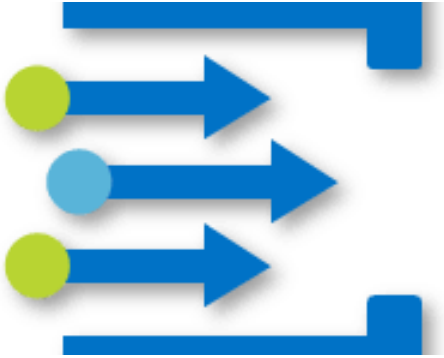
How can we automate this?



Down the rabbit hole 🐰

What is Azure Event Grid?

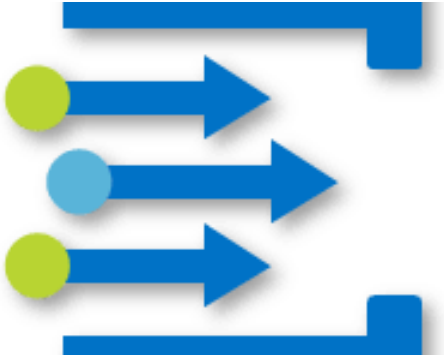
Azure Event Grid is a highly scalable and fully managed **publish-subscribe service for message distribution**. Event Grid offers flexible message consumption patterns and uses the Message Queuing Telemetry Transport (MQTT) and HTTP protocols. With Event Grid, you can build data pipelines with device data, integrate applications, and build event-driven serverless architectures.



Down the rabbit hole 🐰

What is Azure Event Grid?

publish-subscribe service for message distribution.



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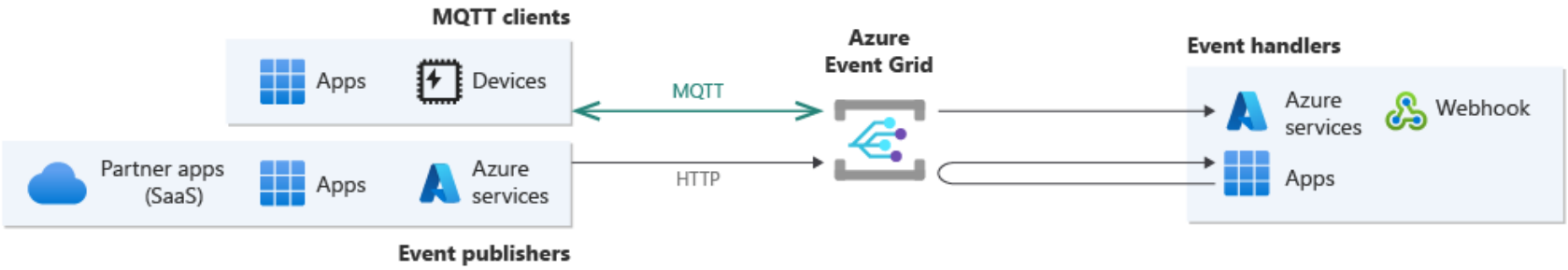

Microsoft Learn (https://learn.microsoft.com/en-us/azure/event-grid/overview)

Down the rabbit hole

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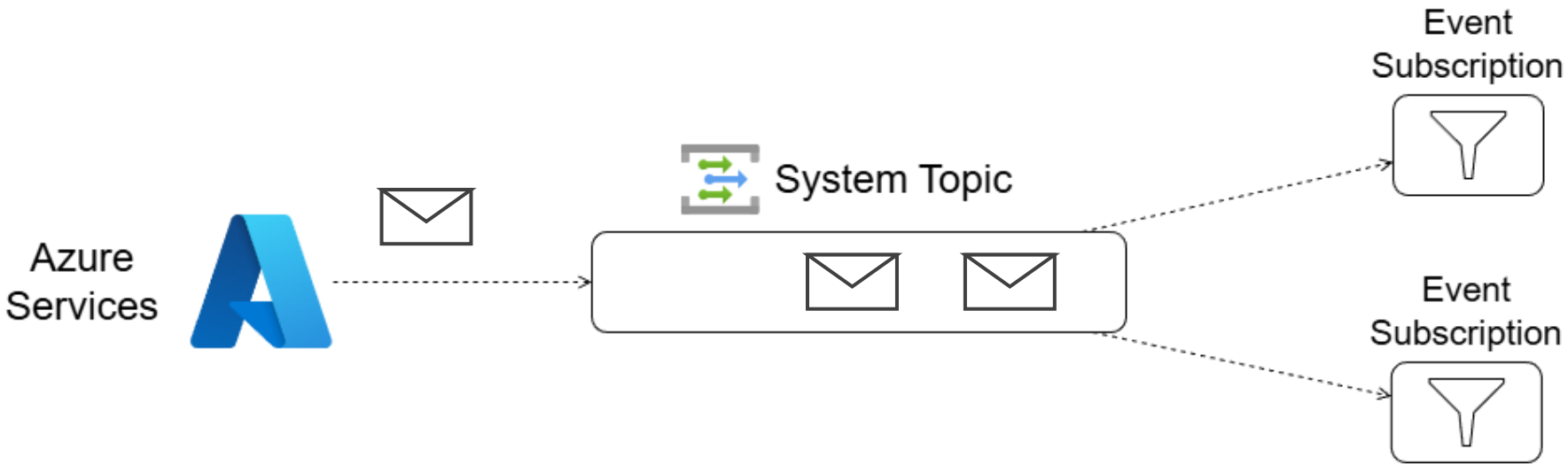

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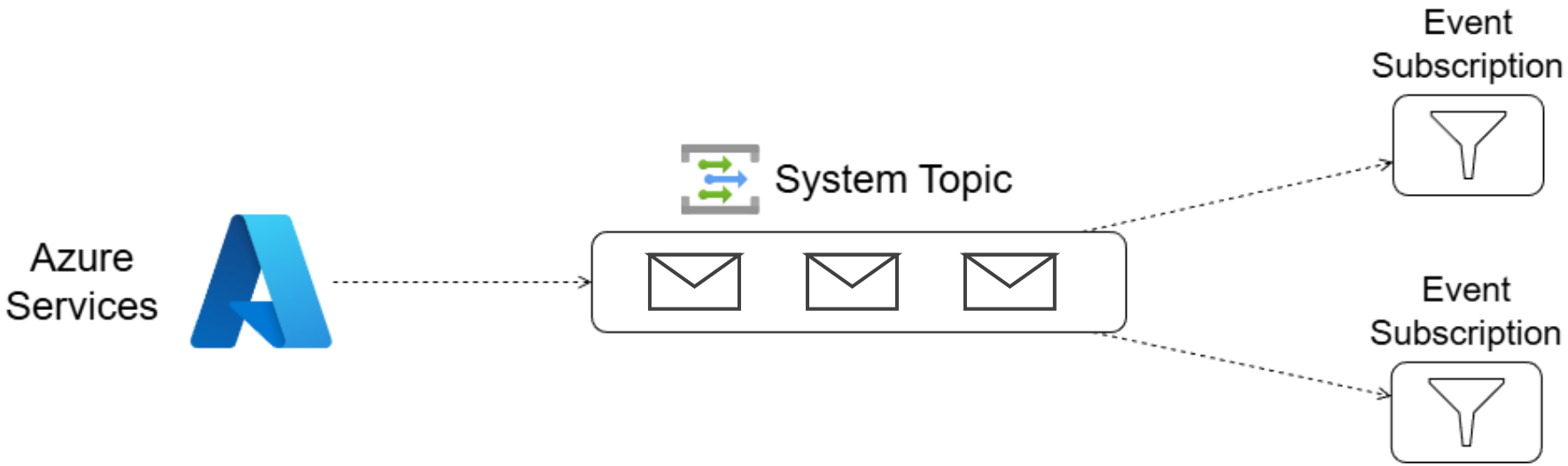

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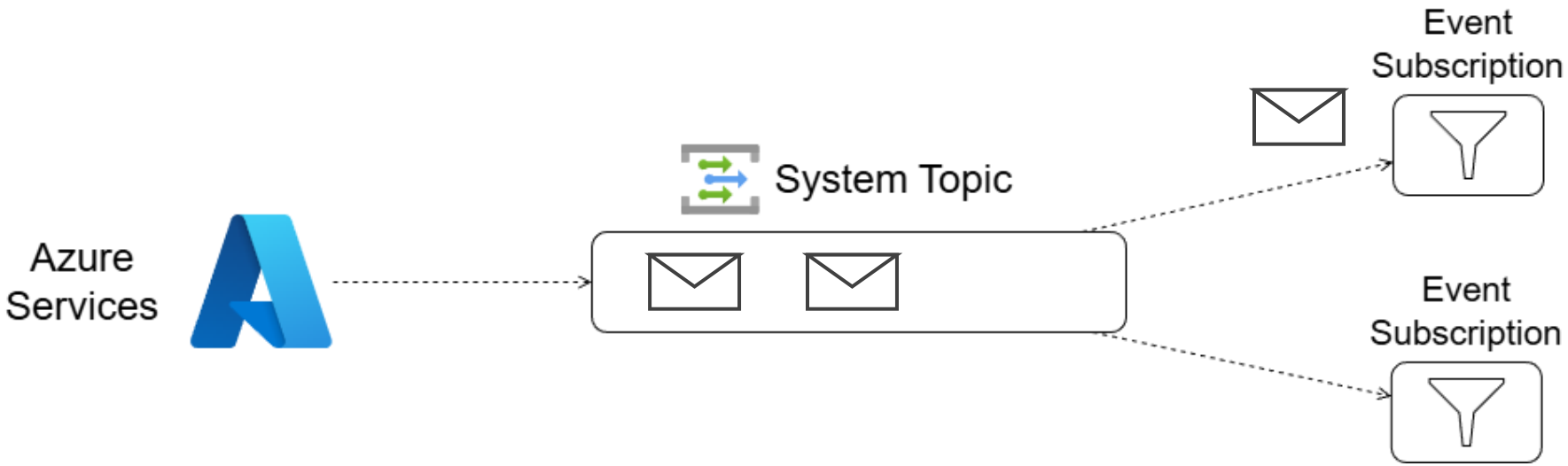

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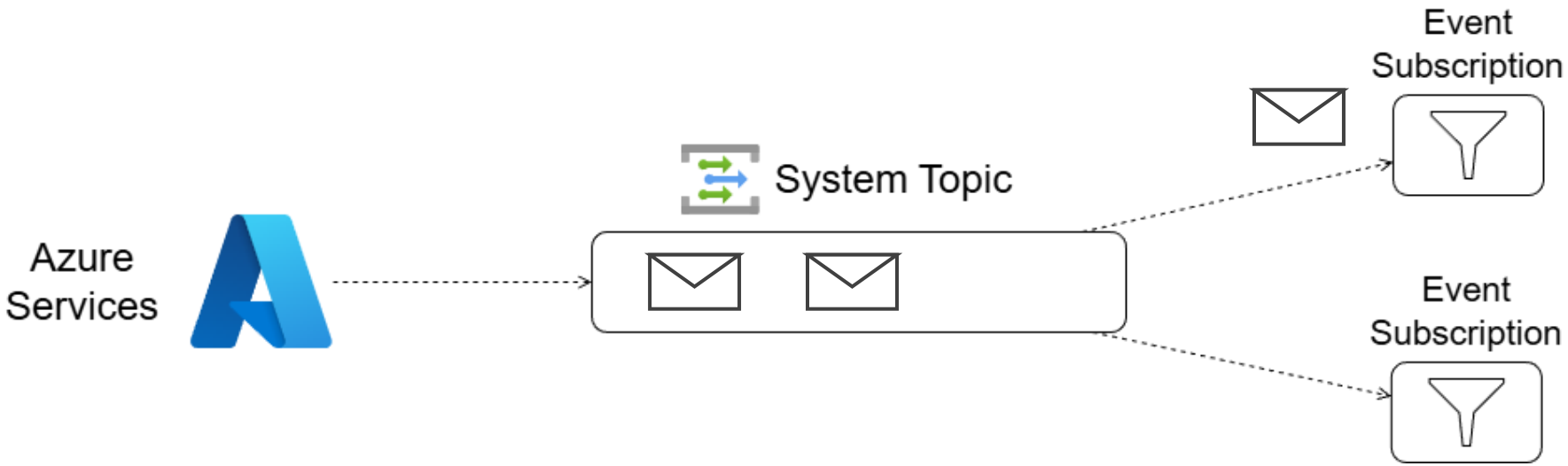

Down the rabbit hole 🐰

What is Azure Event Grid?

- Subscription
- Azure Resource Notifications
- App Service
- Policy
- Blob Storage
- Many more..

Down the rabbit hole 🐰

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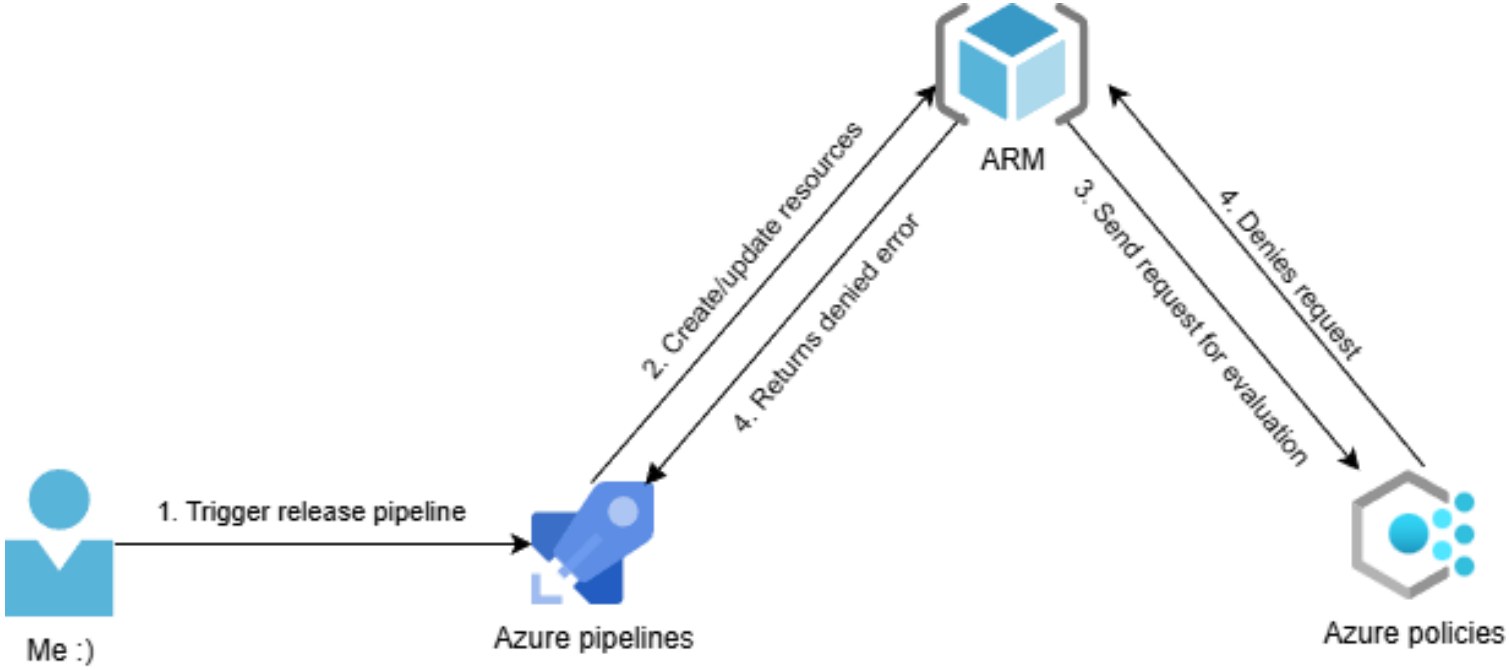
Down the rabbit hole

How can we use Azure Event Grid System Topics?



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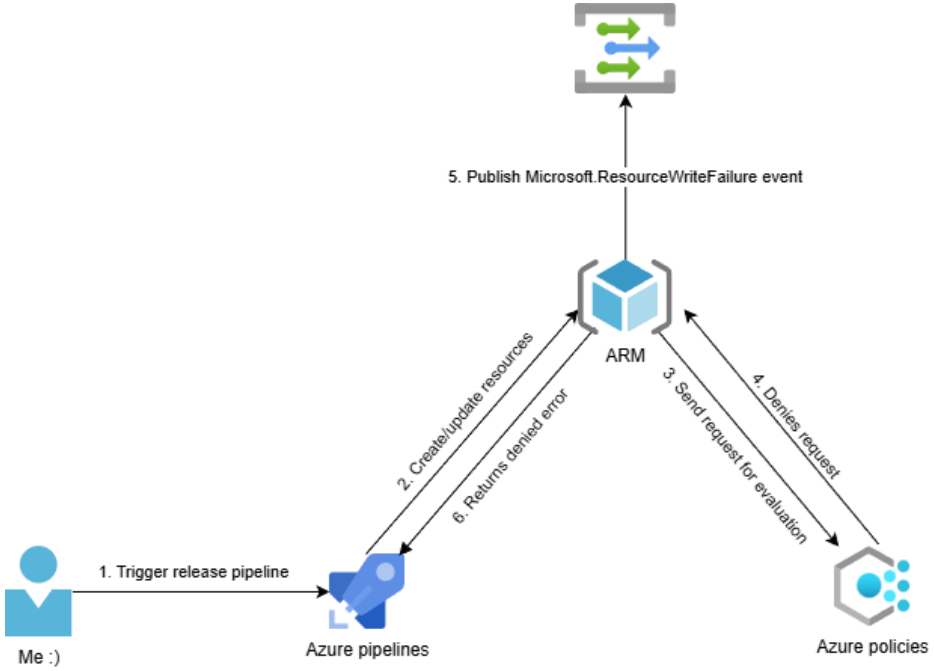

Down the rabbit hole

How can we use Azure Event Grid System Topics?


```
[
    {
        "subject": "/subscriptions/{subscription-id}/resourcegroups/{resource-
group}/providers/Microsoft.Storage/storageAccounts/{storage-name}",
        "topic": "/subscriptions/{subscription-id}",
        "type": "Microsoft.Resources.ResourceWriteSuccess",
        "time": "2018-07-19T18:38:04.6117357Z",
        "id": "4db48cba-50a2-455a-93b4-de41a3b5b7f6",
        "data": {
            "authorization": {
                "scope": "/subscriptions/{subscription-id}/resourcegroups/{resource-
group}/providers/Microsoft.Storage/storageAccounts/{storage-name}",
                "action": "Microsoft.Storage/storageAccounts/write",
                "evidence": {
                    "role": "Subscription Admin"
                }
            },
            "claims": {},
            "correlationId": "{ID}",
            "resourceProvider": "Microsoft.Storage",
            "resourceUri": "/subscriptions/{subscription-id}/resourcegroups/{resource-
group}/providers/Microsoft.Storage/storageAccounts/{storage-name}",
            "operationName": "Microsoft.Storage/storageAccounts/write",
            "status": "Succeeded",
            "subscriptionId": "{subscription-id}",
            "tenantId": "{tenant-id}"
        },
        "specversion": "1.0"
    }
]
```

Down the rabbit hole

How can we use Azure Event Grid System Topics?


```
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        "time": "2018-07-19T18:38:04.6117357Z",
        "id": "4db48cba-50a2-455a-93b4-de41a3b5b7f6",
        "data": {
              "authorization": {
                  "scope": "/subscriptions/{subscription-id}/resourcegroups/{resource-
group}/providers/Microsoft.Storage/storageAccounts/{storage-name}",
                  "action": "Microsoft.Storage/storageAccounts/write",
                  "evidence": {
                      "role": "Subscription Admin"
                  }
              },
            "claims": {},
            "correlationId": "{ID}",
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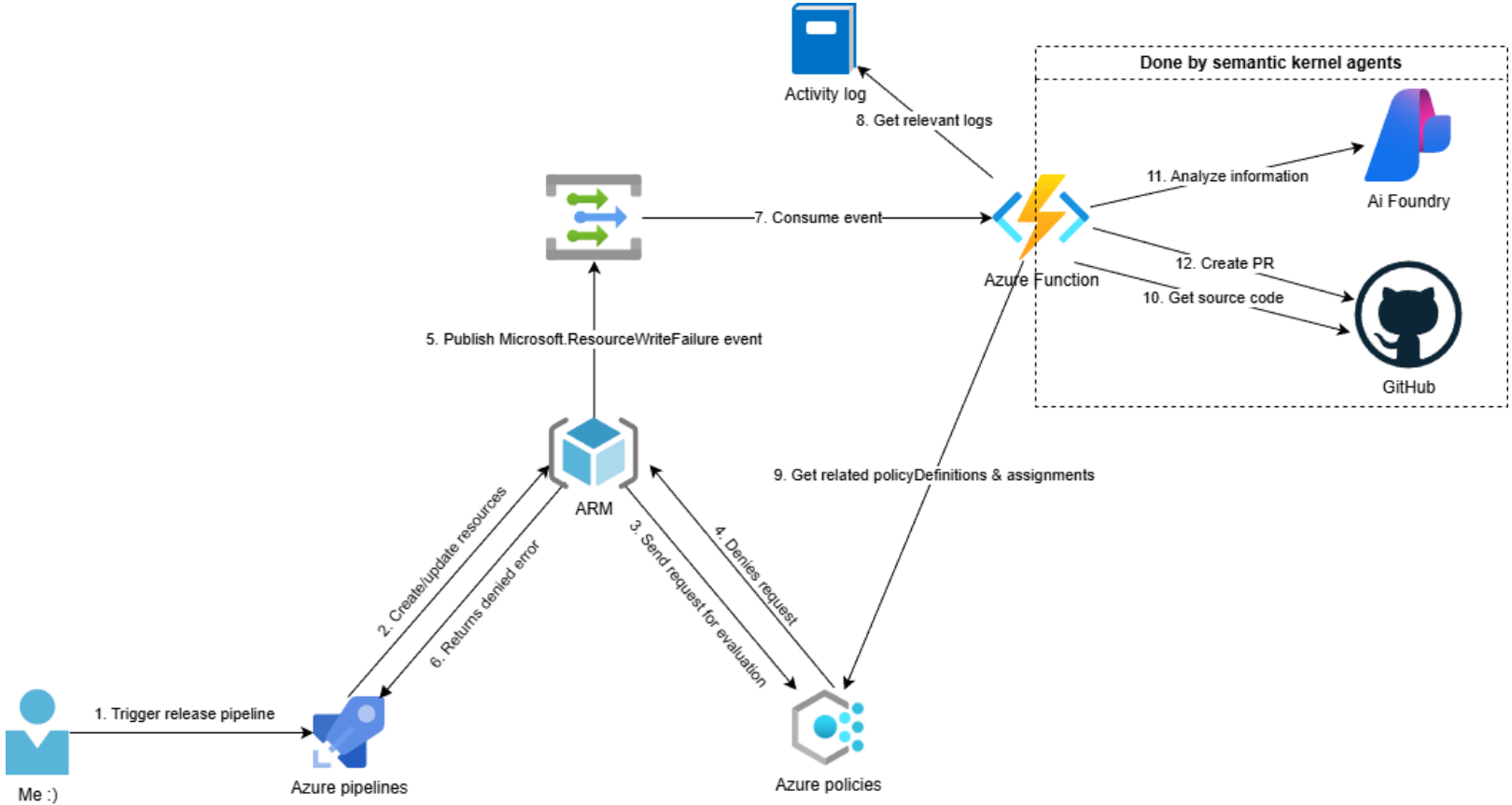

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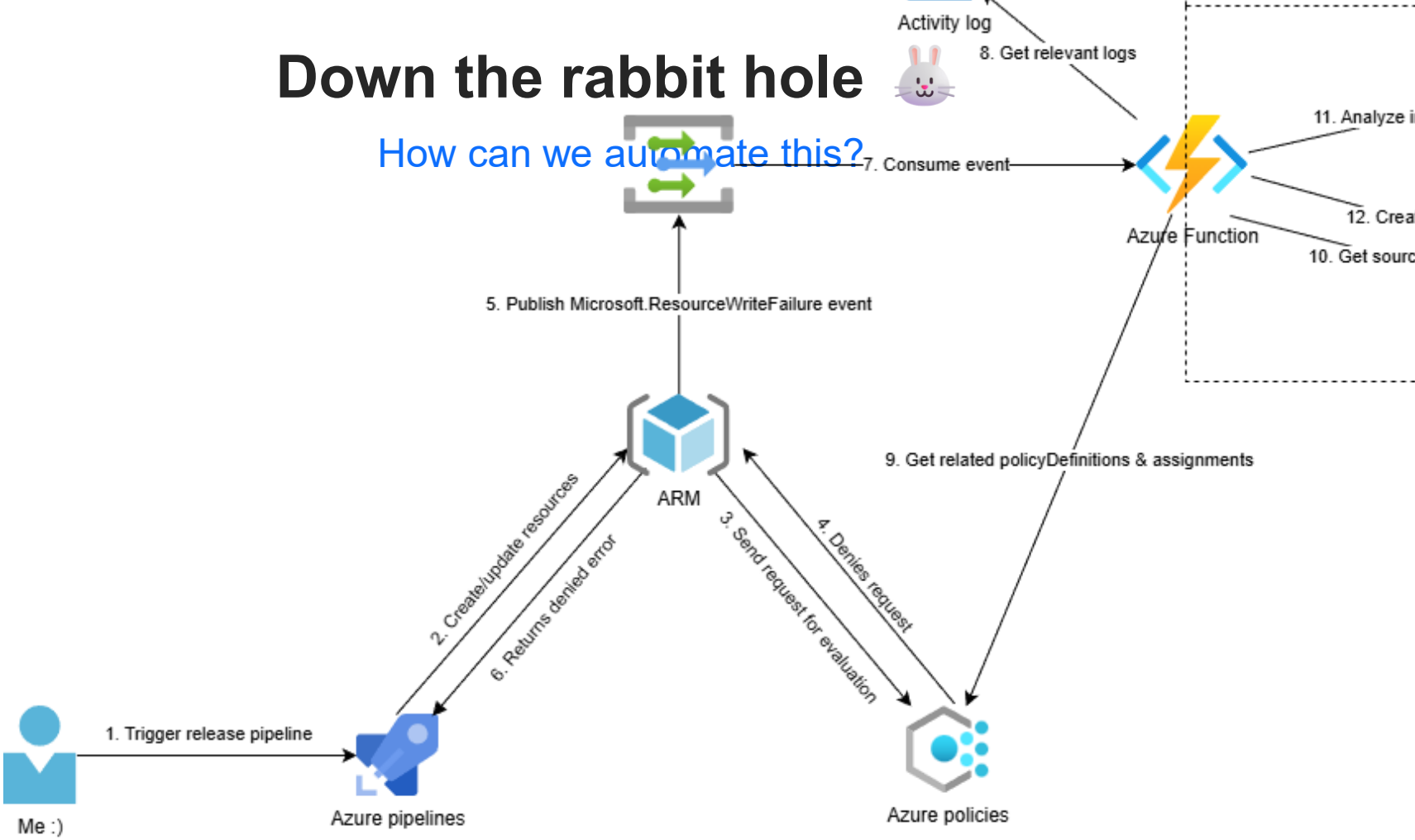

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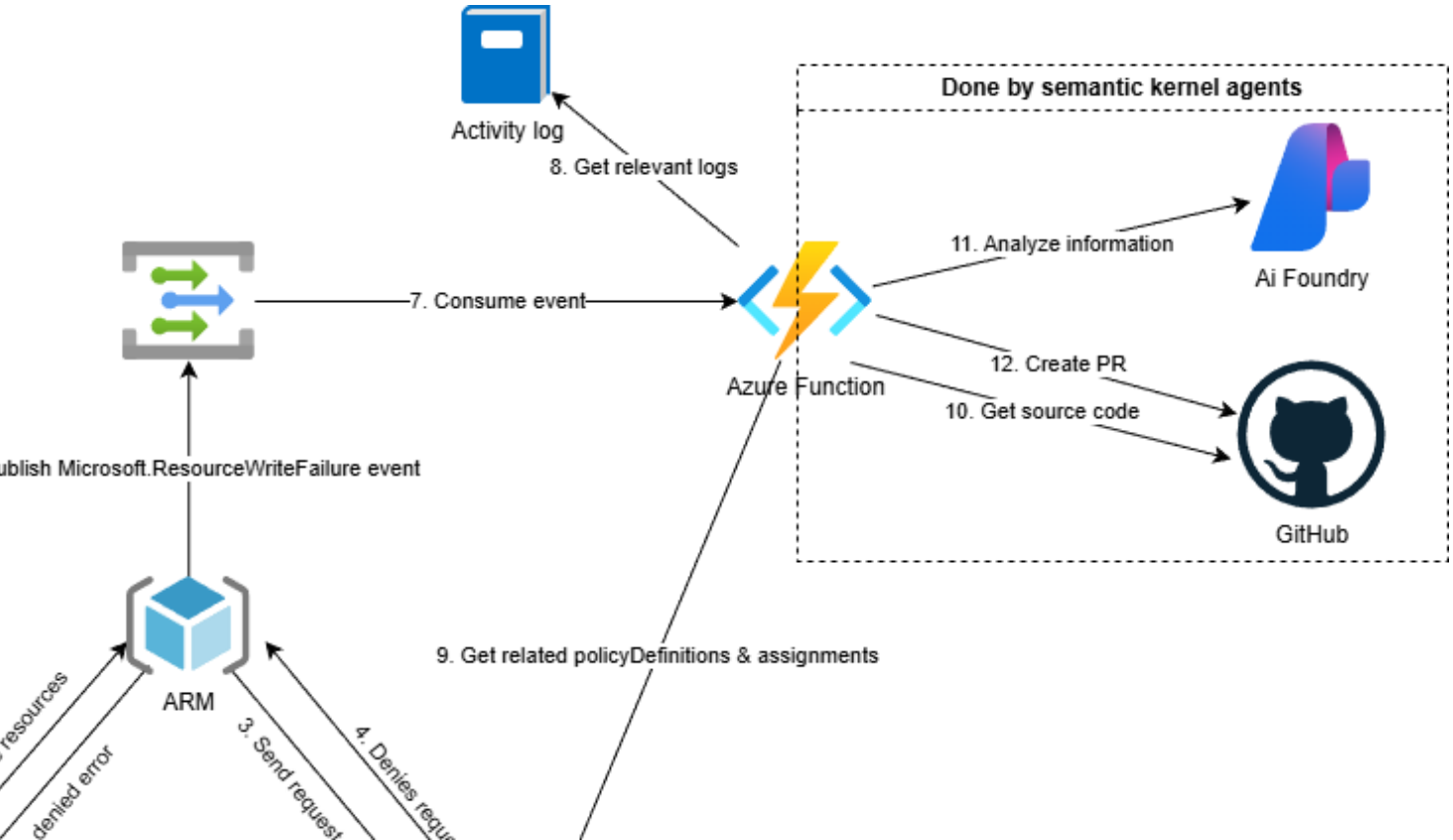

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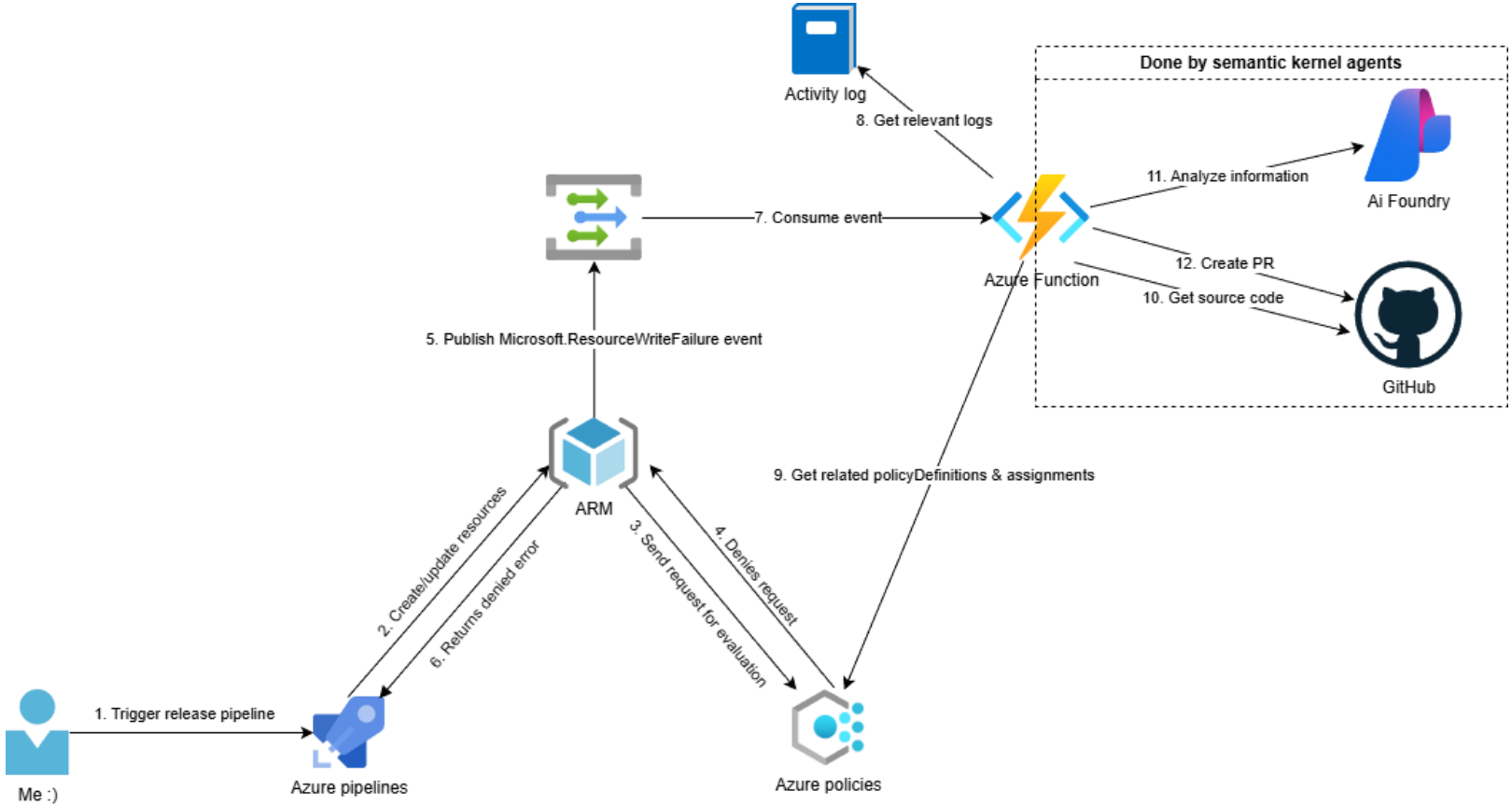

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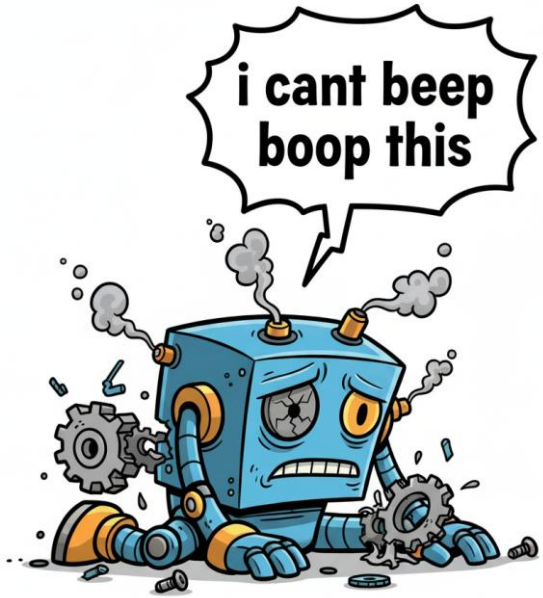

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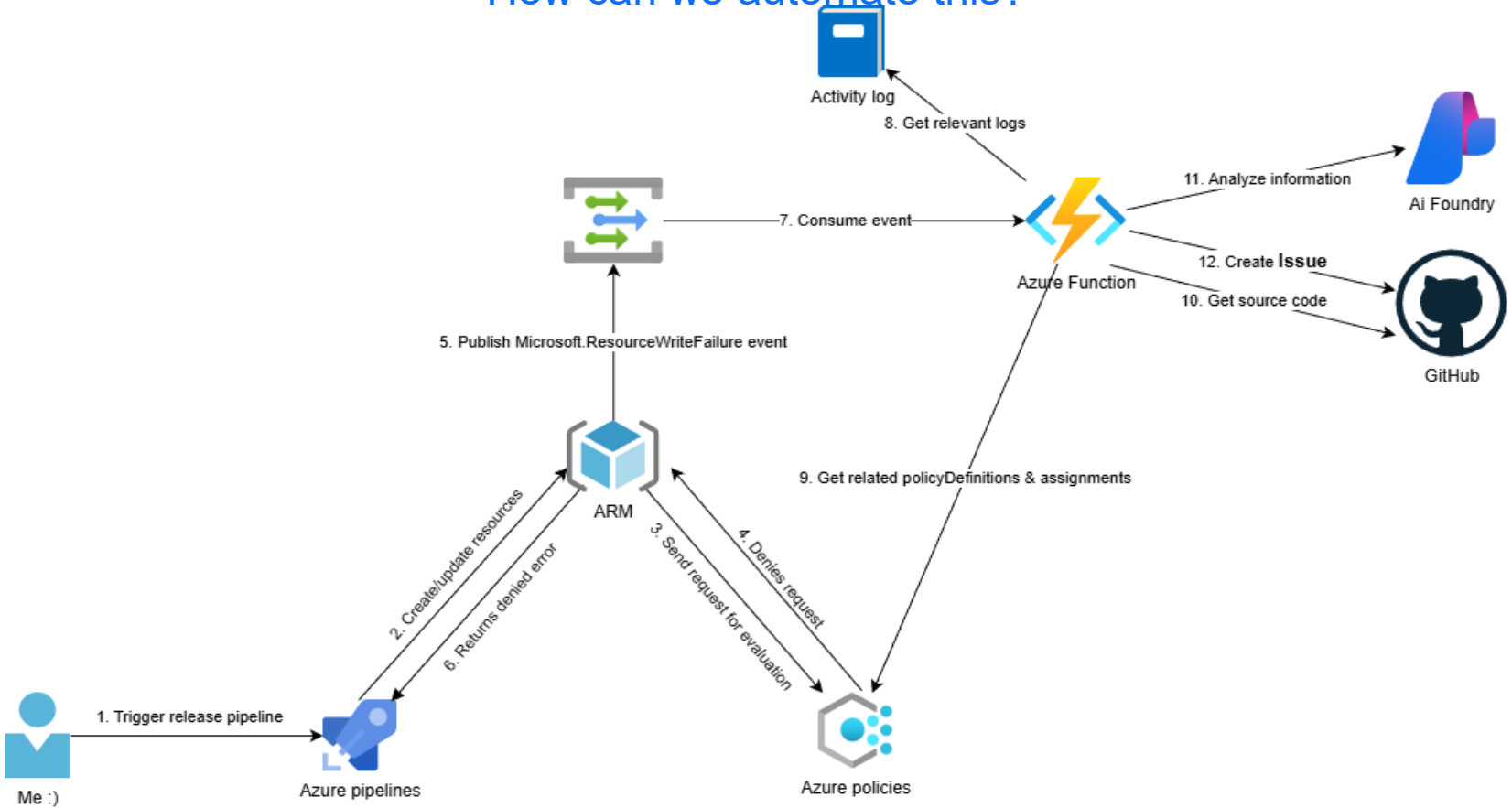

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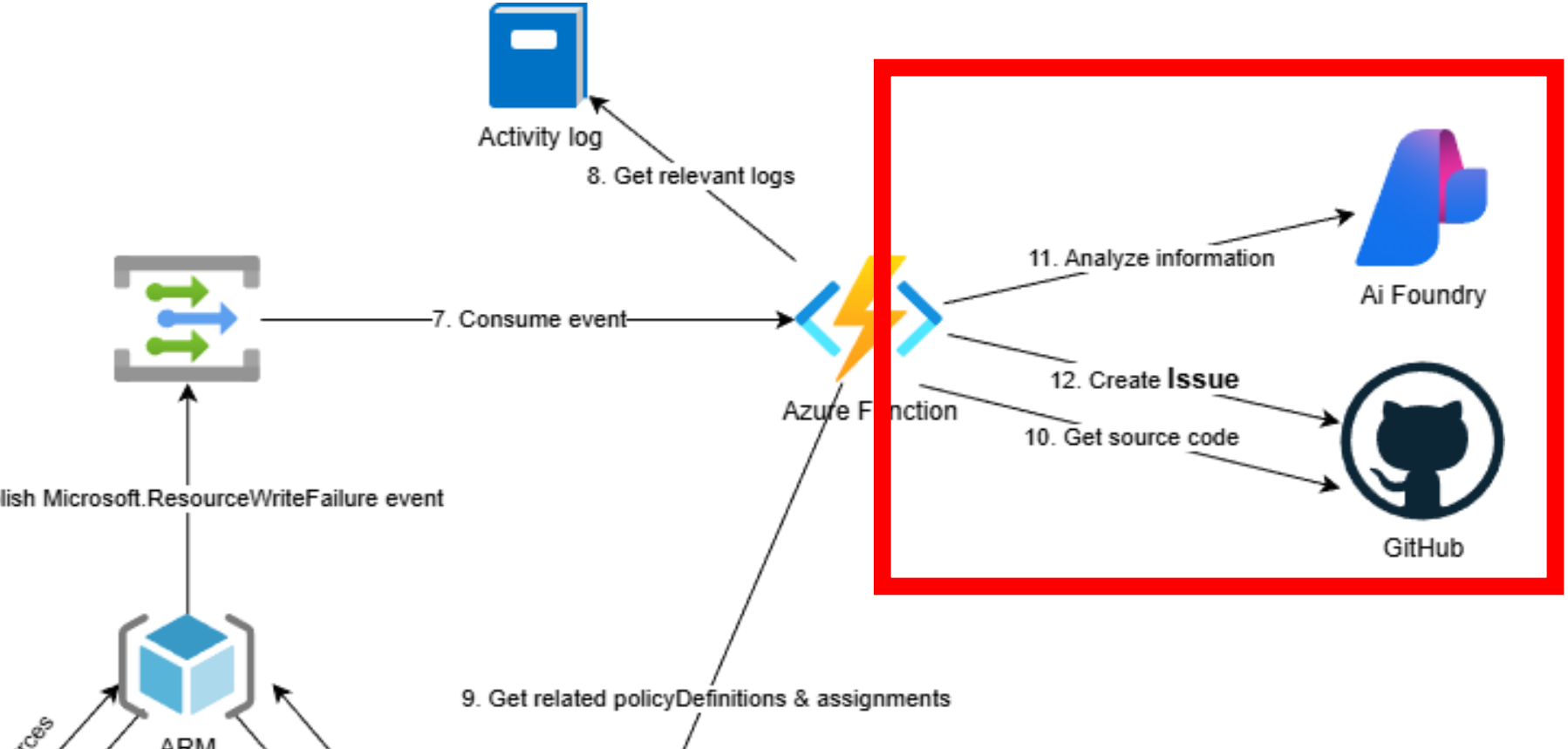
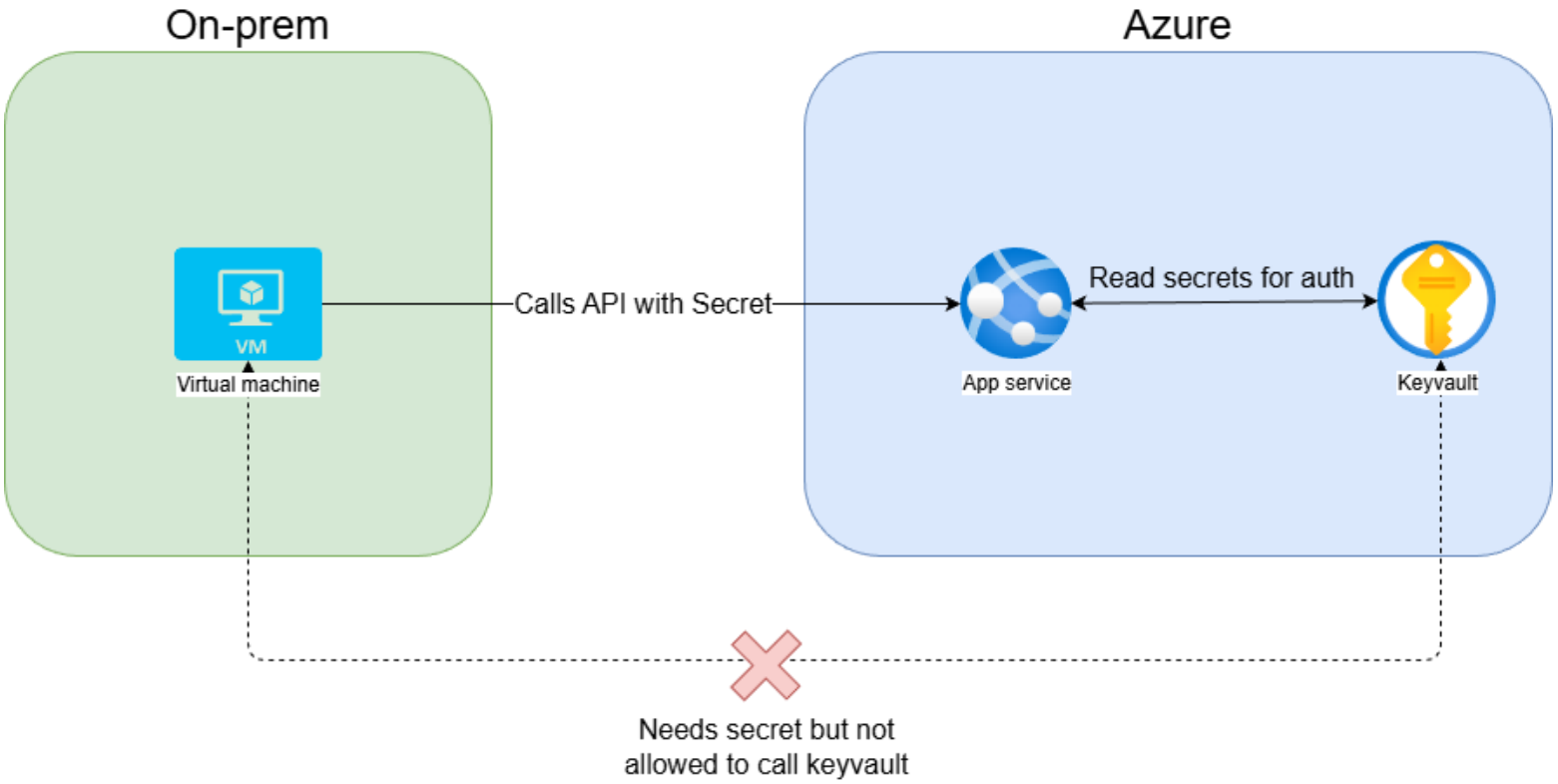


LIVE DEMO

Explore even further

What other possibilities do we have?



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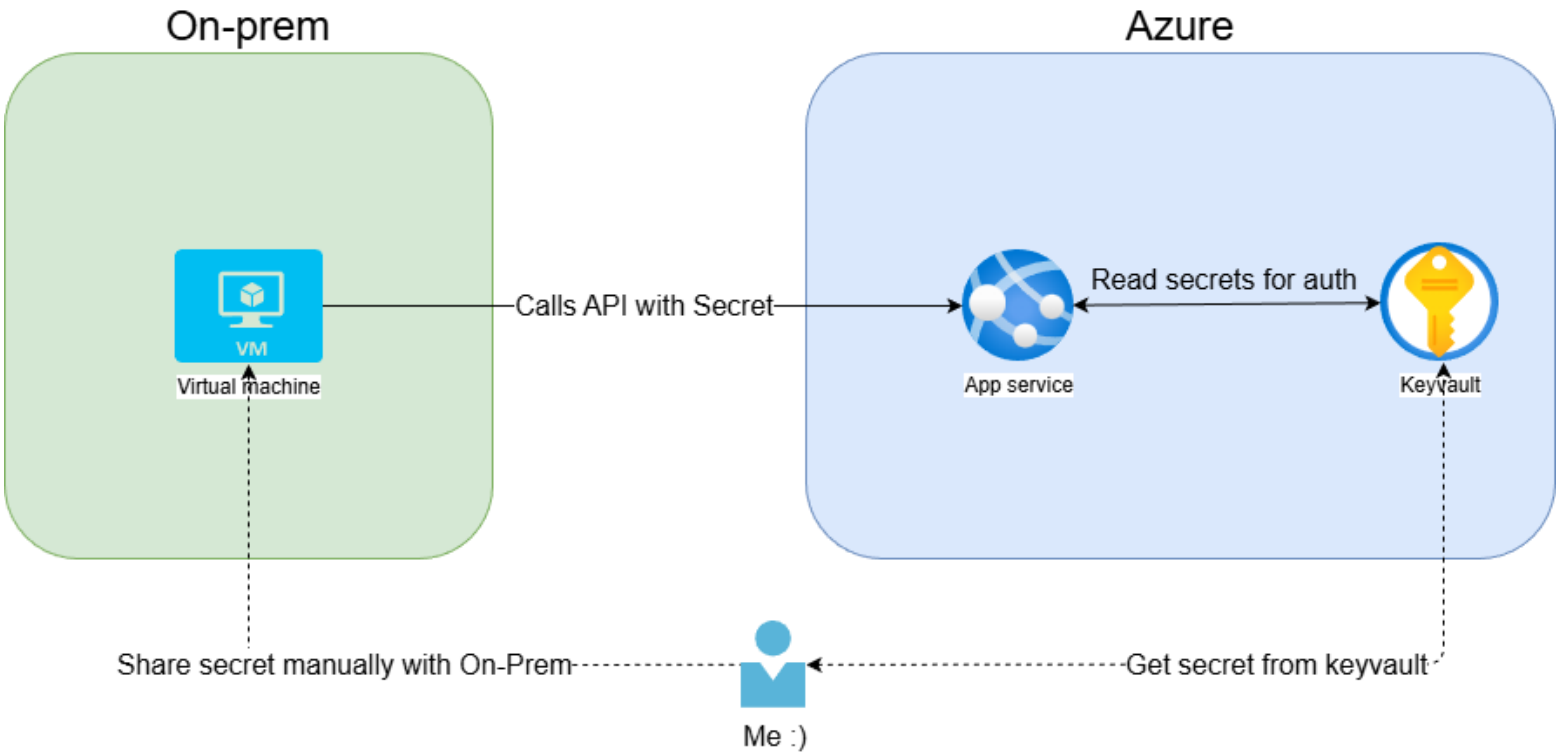

Explore even further 🔭

What other possibilities do we have?

Keyvault as source

Microsoft.KeyVault.SecretNewVersionCreated

Microsoft.KeyVault.SecretNearExpiry

Explore even further

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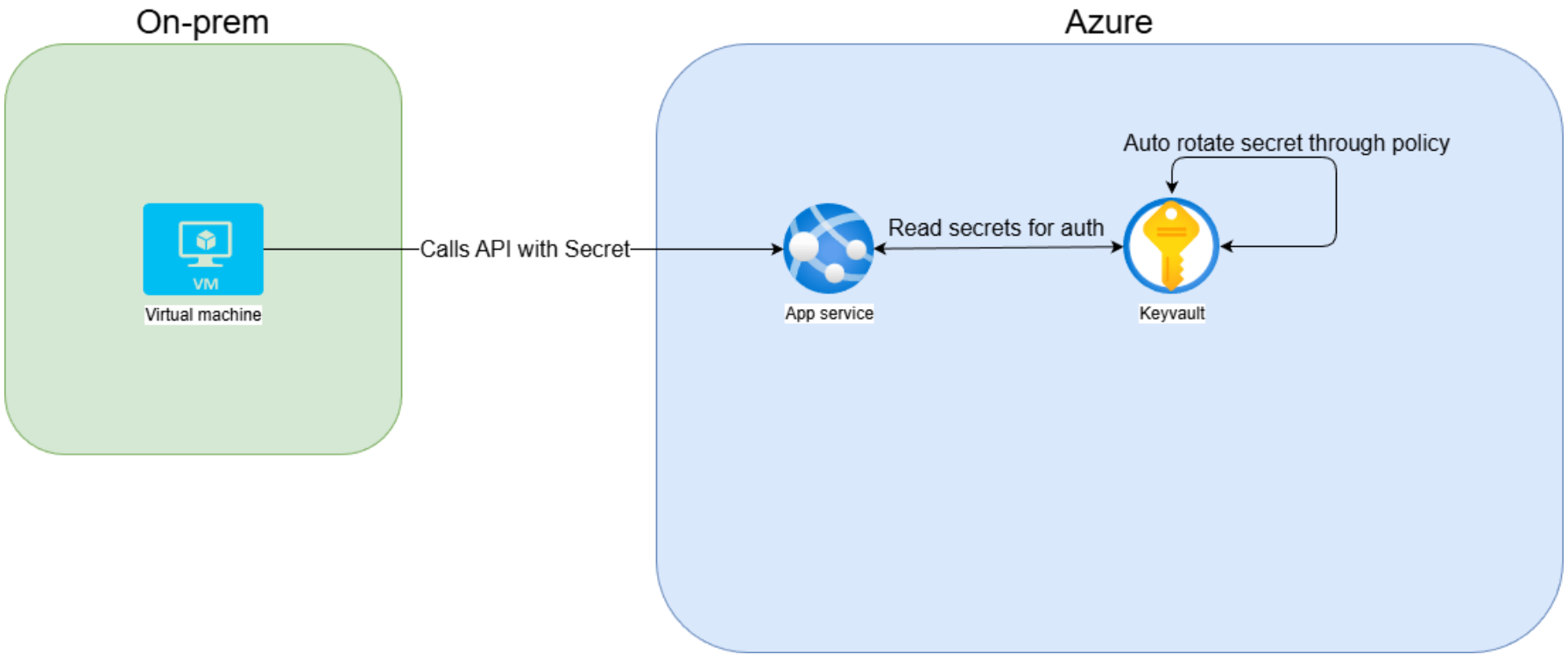

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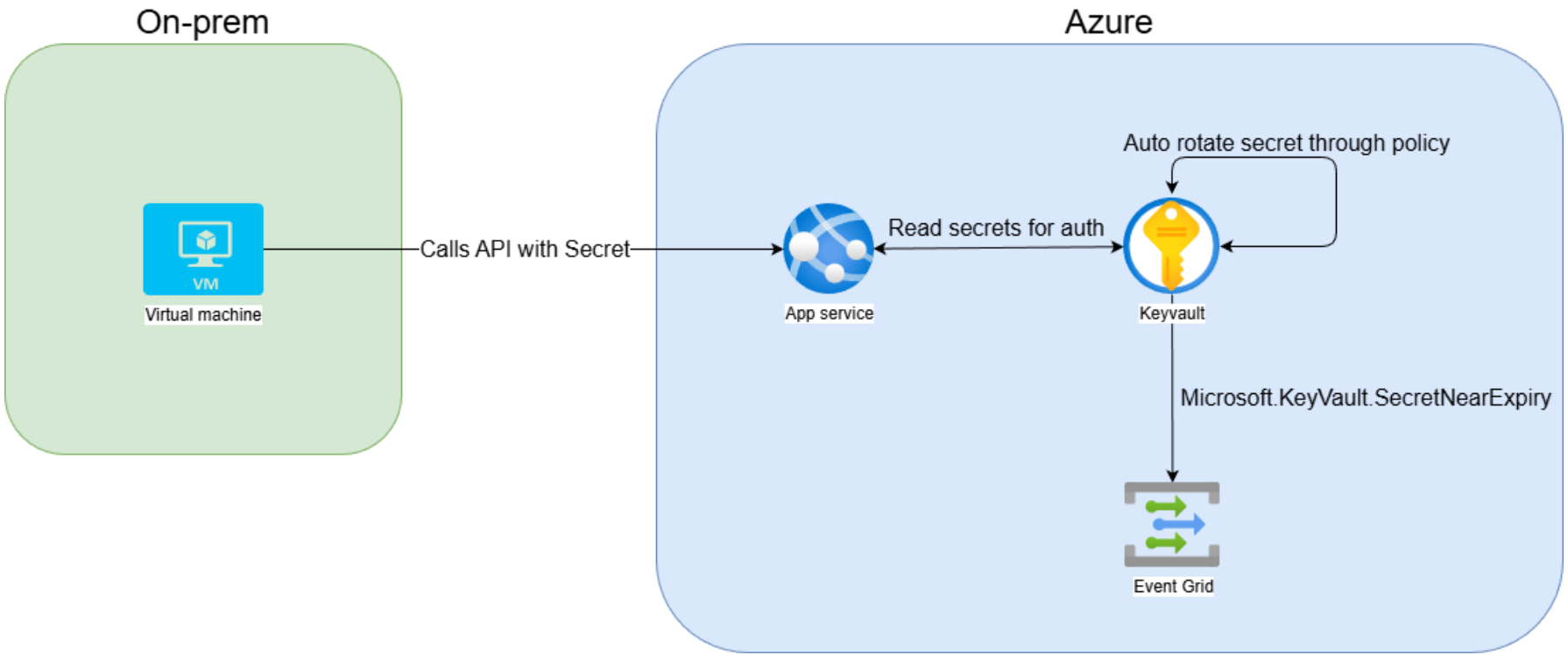

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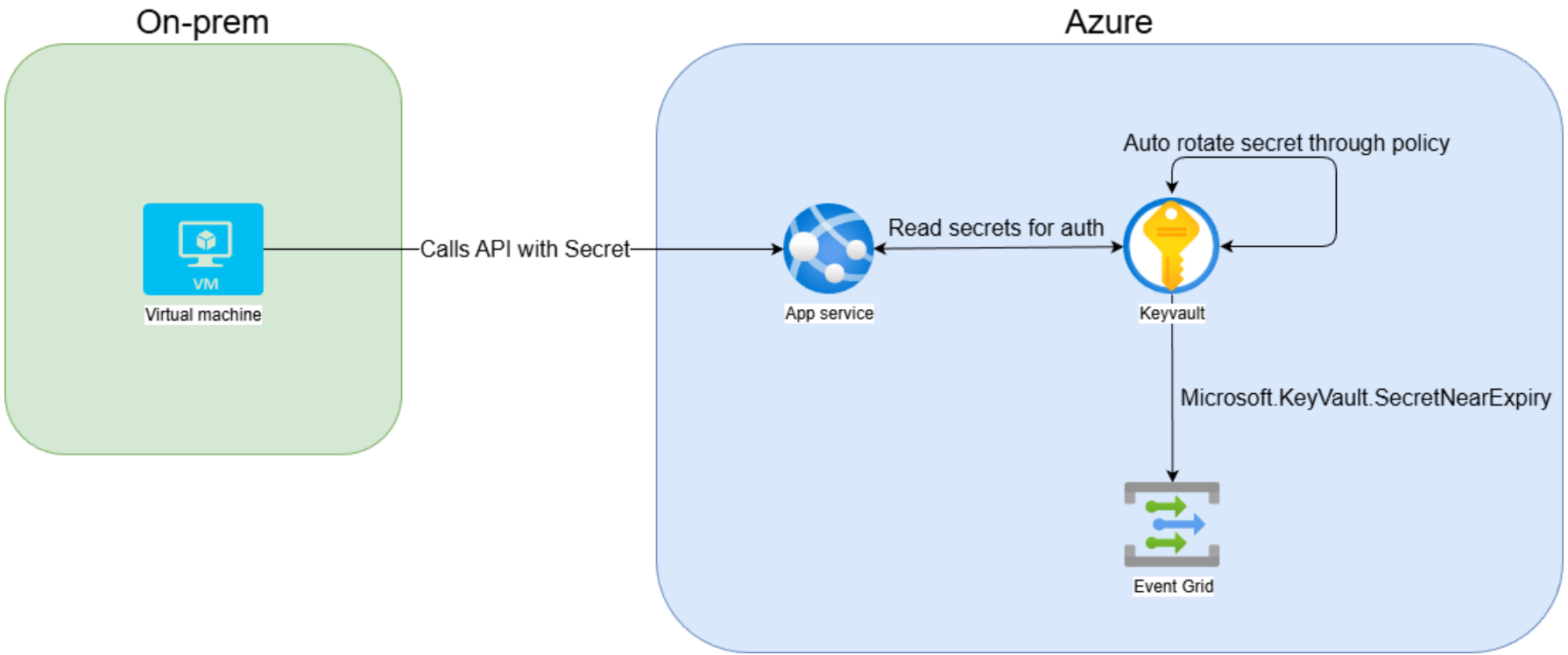

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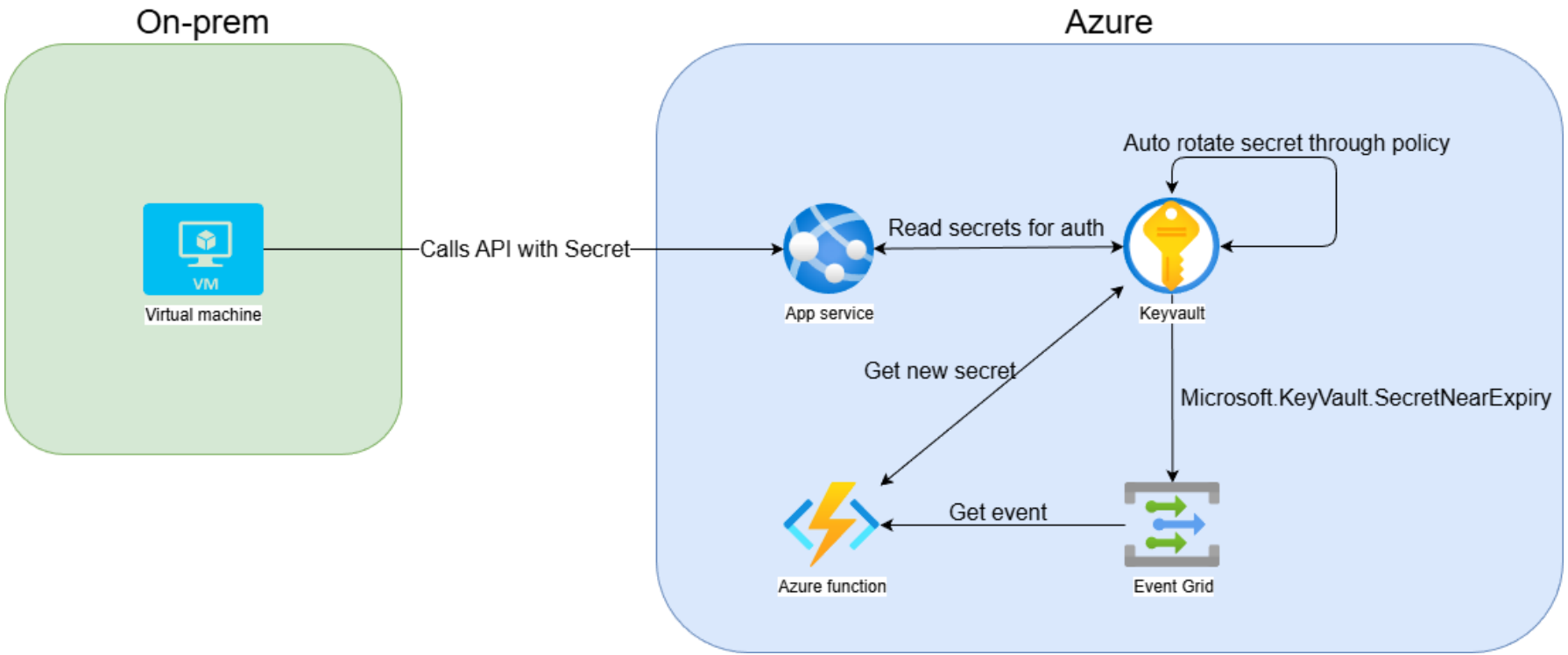

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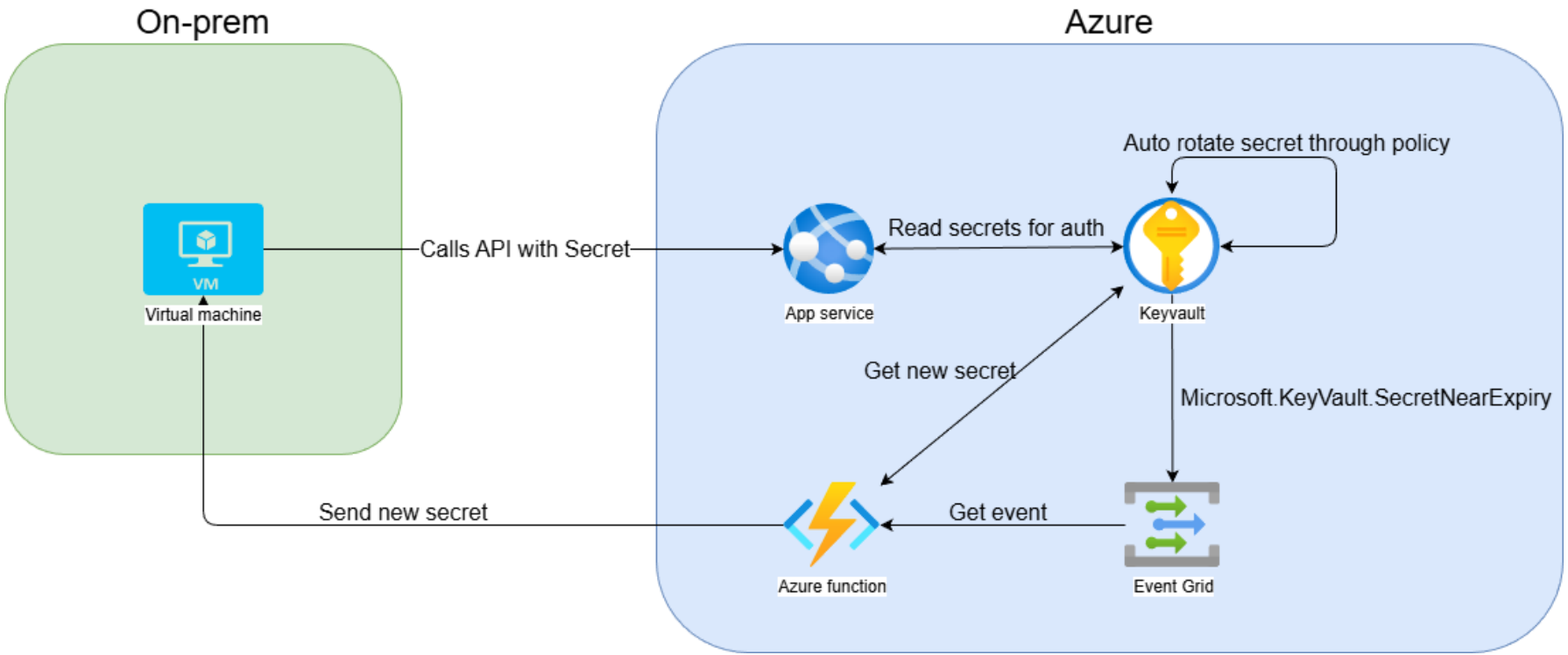

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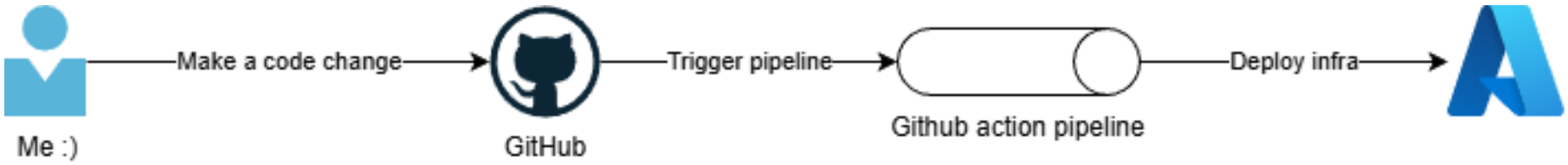

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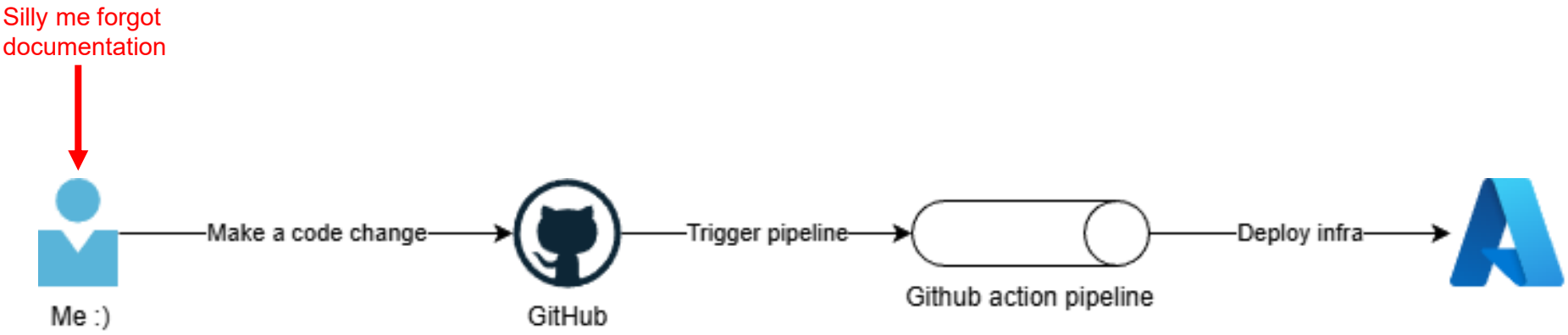

Explore even further 🔭

What other possibilities do we have?

Azure Resource Notifications

- Azure Resource Management (ARM)
- Container Service Event Resources
- Health Resources (Preview)

Explore even further 🔭

What other possibilities do we have?

Azure Resource Notifications

Azure Resource Management (ARM)

Explore even further

What other possibilities do we have?

Azure Resource Notifications
Azure Resource Management (ARM)

- **Provides insights into life cycle of resources**
- **CreateOrUpdated & Deleted Events**
- **Only sends events on success**

Explore even further 🔭

What other possibilities do we have?

```
{
    "id": "string",
    "source": "string",
    "subject": "string",
    "data": {
        "resourceInfo": {
            "id": "string",
            "name": "string",
            "type": "string",
            "location": "string",
            "tags": "string",
            "properties": {
                "_comment": "object-unique-to-each-publisher"
            }
        },
        "apiVersion": "string",
        "operationalInfo": {
            "resourceEventTime": "datetime"
        }
    },
    "type": "string",
    "specversion": "string",
    "time": "string"
}
```

Explore even further 🔭

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             "type": "string",
             "location": "string",
             "tags": "string",
             "properties": {
                 "_comment": "object-unique-to-each-publisher"
             }
         },
        "apiVersion": "string",
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Explore even further

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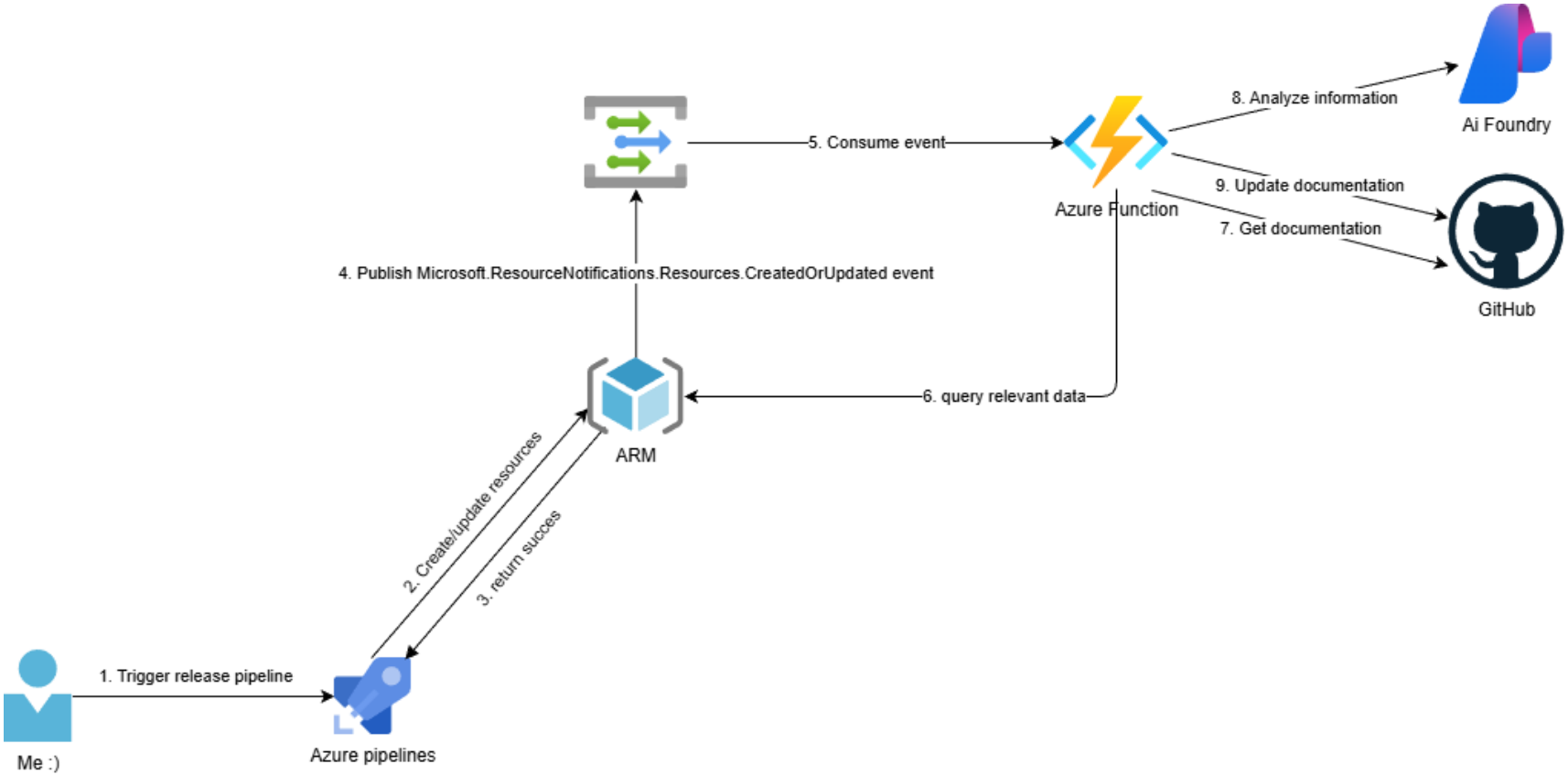

Explore even further 🔭

What other possibilities do we have?

There are so many possibilities!

Go experiment 🧪

How can I properly get started?

1. What processes are not fun or take allot of time?
2. Is it worth the time investment to automate?
3. Is there any event that can be created for this process?

Go experiment 🧪

How can I properly get started?

- Begin small
- Use AI to your advantage
- Use what you know
- If no event exists, you can publish your own..
- Think outside the box
- Have fun! :)

Additional sources

More info in case I forgot :)

- https://codewithhans.com/part-1-design-automate-azure-policy-remediation-directly-in-your-repository-with-event-grid-system-topics
- https://learn.microsoft.com/en-us/azure/event-grid/
- https://learn.microsoft.com/en-us/azure/event-grid/system-topics
- https://learn.microsoft.com/en-us/azure/event-grid/event-schema-resources

Thank You! :)

Questions?

hans@backnook.nl • backnook.nl